Hackathon-to-Hire AI Pre-Assessment Submission

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Graduation Percentage	83%
Passing Year	2026

Section 1 — LLM Project

AI-Based Smart Food Ordering Assistant

Problem Statement:
Restaurants often struggle to manage customer orders efficiently during peak hours. Customers also face difficulty in choosing food items based on preferences, budget, or dietary needs.

Solution:
Developed a Smart Food Ordering System where users can browse menu items, place orders, and receive intelligent food suggestions. The system can be extended using an LLM API for personalized recommendations and conversational assistance.

High-Level Architecture:
• Frontend: HTML, CSS, JavaScript
• Backend: Python / Flask
• Database: MySQL
• AI Integration for NLP-based interaction and recommendation systems

Key Components:
1. User Authentication Module
2. Menu Management System
3. Order Processing Module
4. Database Management
5. AI Recommendation/Chat Module

Engineering Challenge:
Handling different user preferences while maintaining accurate recommendations and ensuring synchronization between frontend and backend during live order updates.

What I Learned:
Backend API integration, database management, AI-assisted workflows, and scalable application architecture.

GitHub Repository:
https://github.com/varshini-1702/srivarshini-s

Demo Video:
N/A

Section 2 — Architecture Design

Goal:
Build a scalable document Question & Answer system for 10,000 internal documents including PDFs, DOCX files, and Confluence pages.

Document Ingestion:
Load and process PDFs, Word files, and Confluence documents using document loaders and APIs.

Chunking Strategy:
Use Recursive Character Text Splitting with chunk sizes between 500–1000 tokens and overlaps of 100–150 tokens to preserve semantic meaning.

Embedding Model Choice:
• OpenAI text-embedding-3-small
• Sentence Transformers (all-MiniLM-L6-v2)

Vector Store:
Use ChromaDB or Pinecone for scalable vector similarity search and metadata filtering.

Retrieval Strategy:
1. Semantic Vector Search
2. Hybrid Retrieval (Keyword + Semantic Search)
3. Top-K Retrieval
4. Re-ranking for improved relevance

Handling Partial Answers:
Retrieve multiple relevant chunks, merge contextual information, provide citations, and request clarification when confidence is low.

Biggest Failure Mode:
Hallucinated responses due to weak or incomplete retrieval context.

Mitigation:
Strict context grounding, confidence scoring, retrieval validation, citation support, and human feedback mechanisms.

Section 3 — Debug Challenge

INSUFFICIENT INFORMATION PROVIDED. This assessment is intended exclusively for evaluating a candidate’s technical and problem-solving abilities. The code and context supplied are deliberately incomplete and are not intended to be solved using AI-assisted tools or autogenerated responses. Any AI-generated solution may be inaccurate, misleading, or considered a violation of the assessment guidelines.
